

# Uncertainty intervals

Project: Reproducible AI Research Study

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of uncertainty intervals.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of uncertainty intervals

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Sampling variability. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

### My starting explanation

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### This week's end goal

Explain uncertainty intervals using the records and boundaries of this fictional investigation.

## 2. Notice the evidence

An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus  $1.96 \cdot \sigma / \sqrt{n}$ . Unknown sigma requires a different justified method.

### Uncertainty intervals — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Observed mean 10,  $n=16$ ,  $\sigma=2$ . Compute standard error, margin and interval.

**Check or calculation:** For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus  $1.96 \cdot \sigma / \sqrt{n}$ . Unknown sigma requires a different justified method.

**Conclusion and limit:**  $SE=2/4=.5$ ; margin=.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in 95% of repeated samples.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

Observed mean10, n16, sigma2. Compute standard error, margin and interval. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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### Change one thing in the pictured case

Change: Use a supplied interval  $[-1.4, 0.3]$  instead.

Prediction: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.

Reason: The new upper endpoint crosses zero.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Standard error

**Meaning:** The model-based variation of an estimate

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**Word:** Confidence interval

**Meaning:** An interval procedure with declared repeated-sampling coverage

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**Word:** Known sigma

**Meaning:** A supplied population standard deviation assumption

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

Observed mean 10,  $n=16$ ,  $\sigma^2=4$ . Compute standard error, margin and interval.

$SE=2/4=.5$ ;  $\text{margin}=.98$ ;  $\text{interval}[9.02, 10.98]$ . Under assumptions, the method covers the fixed population mean in 95% of repeated samples.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
from math import sqrt

def known_sigma_interval(mean, sigma, n, critical=1.96):

    """Assume independent normal sampling and a known population sigma."""

    if type(n) is not int or n < 1 or sigma < 0 or (critical <= 0):

        raise ValueError('interval settings')

    se = sigma / sqrt(n)

    margin = critical * se
```

```
return (se, mean - margin, mean + margin)

print(known_sigma_interval(10, 2, 16))
```

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Uncertainty intervals

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

### The facts we will use

Estimated difference B–A is -0.8 seconds; supplied 95% confidence interval is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

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### Supplied interval for B minus A time

|                            | Lower endpoint (s) | Estimate (s)  | Upper endpoint (s) |
|----------------------------|--------------------|---------------|--------------------|
| Original supplied interval | -1.4               | -0.8          | -0.2               |
| Changed supplied interval  | -1.4               | not specified | 0.3                |

Read a cell using both its row and column labels.

# Uncertainty intervals

## Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Locate the estimate

-0.8 lies between the interval endpoints.



### 2. Interpret the sign

Negative  $B-A$  means  $B$  took less time under this measurement convention.



### 3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

## CHANGE ONE THING

Use a supplied interval  $[-1.4, 0.3]$  instead.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Locate the estimate

**What happens and why:** -0.8 lies between the interval endpoints.

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**Step:** 2. Interpret the sign

**What happens and why:** Negative  $B-A$  means  $B$  took less time under this measurement convention.

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### Step: 3. Respect coverage meaning

**What happens and why:** 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

### What this example does not establish

An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

## 7. Find and repair a mistake

A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Uncertainty intervals”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

**Plan · Do · Check**

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Uncertainty intervals. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

Observed mean10, n16, sigma2. Compute standard error, margin and interval. Show the intermediate steps and give the conclusion with its scope.

```
from math import sqrt

def known_sigma_interval(mean, sigma, n, critical=1.96):
    """Assume independent normal sampling and a known population sigma."""
    if type(n) is not int or n < 1 or sigma < 0 or (critical <= 0):
        raise ValueError('interval settings')

    se = sigma / sqrt(n)

    margin = critical * se
```

```
return (se, mean - margin, mean + margin)

print(known_sigma_interval(10, 2, 16))
```

### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_  
\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A narrow interval can be precisely biased; sampling independence and

representativeness must be justified. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled uncertainty intervals evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of uncertainty intervals. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_  
\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_  
\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_  
\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_  
\_\_\_\_\_

## Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

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## 13. Study a complete case

**Complete evidence record**

**Uncertainty intervals — evidence trace**

**Input and scope · Check or calculation · Conclusion and limit**

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**Check or calculation:** For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus  $1.96 \cdot \sigma / \sqrt{n}$ . Unknown sigma requires a different justified method.

**Conclusion and limit:**  $SE=2/4=.5$ ; margin=.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in 95% of repeated samples.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

#### My complete trace of the worked case

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## 14. Try a new case independently

### Independent case: Apply a new case independently

Mean 20,  $n=25$ , known  $\sigma=5$  and critical 1.96. Compute interval and name one assumption.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

Mean 20,  $n=25$ , known  $\sigma=5$  and critical 1.96. Compute interval and name one assumption.

#### My independent procedure, result and limitation

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.